

Homework 05

Consider the bar shown in Fig. 1, where the cross-sectional area $A=1\text{ in}^2$, and Young's modulus $E=2.3\times 10^6\text{ psi}$. If $q_1=0.05\text{ in}$ and $q_2=0.06\text{ in}$, determine the following (by hand calculation):

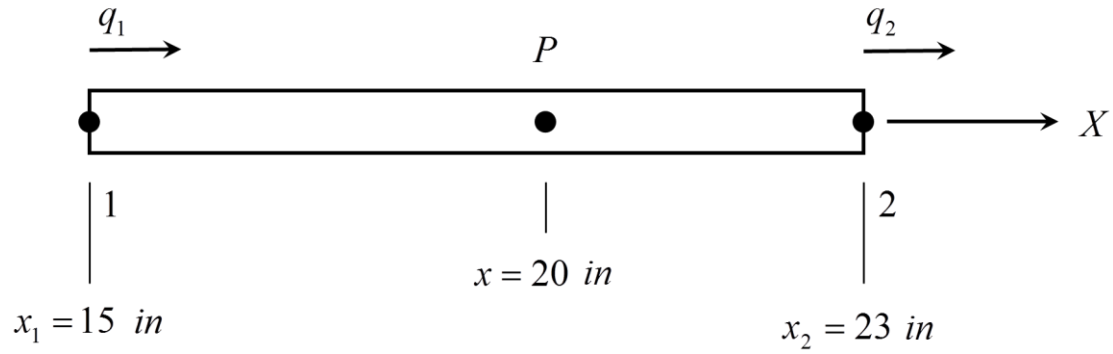


Fig. 1

- (a) the displacement at point P ,
- (b) the strain ϵ and stress σ at point P ,
- (c) the element stiffness matrix,
- (d) the strain energy in the element.